

1 Co-located measurements (same population, place, period)

Drinking water

C_w (µg/L)
IR_w (L/d)
treated-water basis
<LOD: LOD/2 or ROS

Diet (k food groups)

C_k (µg/kg wet wt, as eaten)
IR_k (kg/d)
S_k toxic-species fraction
RBA_k relative bioavailability

Ambient air

C_a (ng/m³, PM10)
ET, EF time-activity
RfC / IUR basis
(RAGS Part F)

2 Route doses on a common scale

$$D_w = C_w \cdot IR_w \cdot EF \cdot ED / (BW \cdot AT)$$

$$D_f = \sum_k C_k \cdot S_k \cdot RBA_k \cdot IR_k \cdot EF \cdot ED / (BW \cdot AT)$$

$$EC_a = C_a \cdot ET \cdot EF \cdot ED / AT$$

population-specific BW, IR, EF, ED; joint distribution sampled by Monte Carlo with inter-media correlation ρ

3 One aggregate index per metal, chosen by mode of action

Threshold agents

HI_{agg} = D_w/RfD_w + D_f/RfD_f
+ EC_a/RfC
Cd, Cr, Cu, Hg, Ni, iAs
flag when HI_{agg} > 1

Non-threshold agents

MOE = BMDL / (D_w + D_f + D_{a,eq})
Pb (BMDL01/10), iAs (BMDL05)
concern when MOE ≤ 1
(≥ 10 reassuring: SBP, CKD)

Carcinogens

ILCR = (D_w + D_f)·CSF
+ EC_a·IUR
iAs, Cr(VI), Cd/Ni (inh.), Pb
lifetime AT; ADAF for Cr(VI)

4 Outputs

- empirical source contribution $f_m = HQ_m / HI_{agg}$ (replaces the default 20 % RSC)
- context-specific water value $C^*_w = RfD \cdot BW \cdot P / IR_w$, $P = \text{clamp}(1 - D_{other}/RfD, 0.2, 0.8)$
- target-organ HI across metals; P50 / P95 of the aggregate (not the sum of route-specific P95s)